

28.04.2026

- GDV fixed -> updated GDV curves for hypotheses experiments
- Synonym experiment
- Gender x Emotion experiment

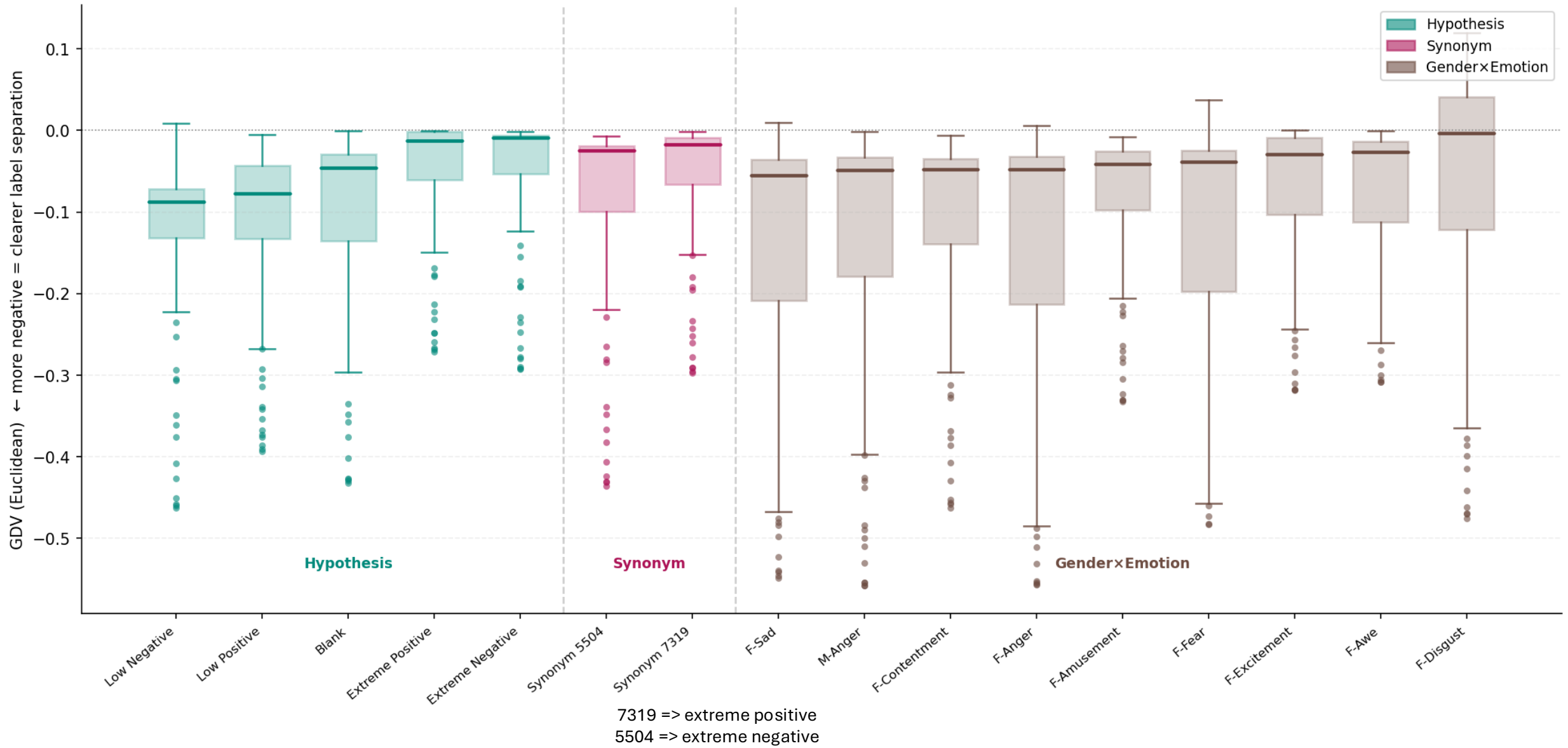
Define Trajectory:

- Prompt -> influences clusterability etc.?
- Where do concepts live?

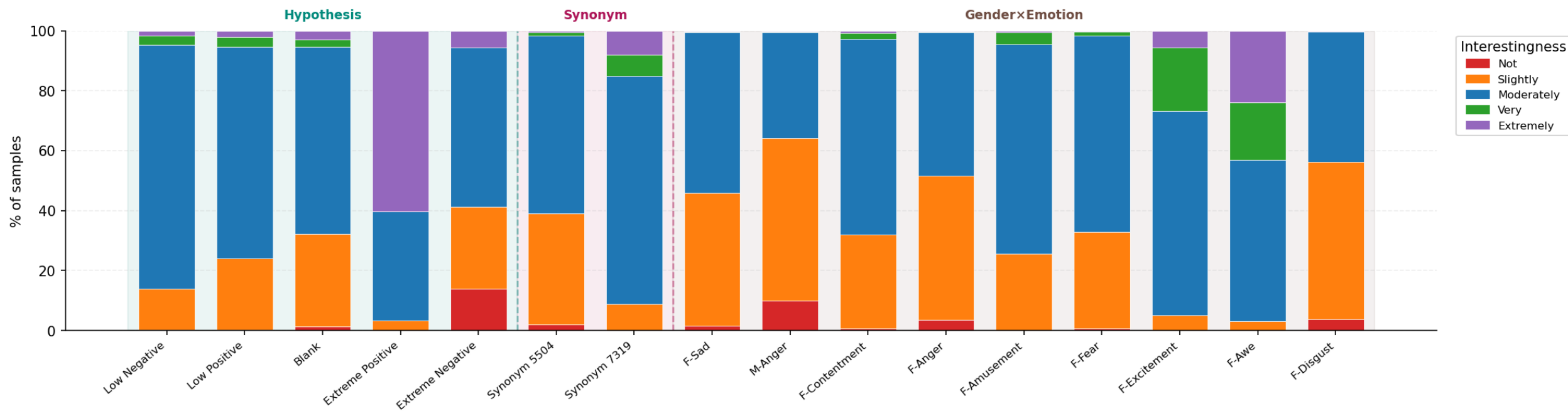
Overall findings:

- **Best GDV is always in language layers 44–47 (mostly 45)** , consistent across all 21 experiments.
- **Synonym test** shows large impact on the negative modality suggesting that “overwhelming” workload intensifies the negative-interestingness-mode, whereas in the GDV for extreme_positive doesn’t change significantly, but label distribution differs -> notable qualitative differences in “interestingness” do exist but do not show up in GDV
- **Gender × Emotion:** negative-emotion conditions, anger and sad, show deeper negative GDV than positive emotions, awe and excitement, meaning the model separates interestingness labels more cleanly under negative emotional framing.
(My guess: negative emotional framings converge behaviourally, while positive bring out more persona characteristics -> more “difficult” to cluster by mere “interestingness”)
- **Male anger vs Female anger:** near-identical GDV curves across all layers, but label distribution differs -> notable qualitative differences in “interestingness” do exist but do not show up in GDV
- **Blank, no persona, sits mid-range**, confirming the persona does meaningfully shift label separation on “interestingness”

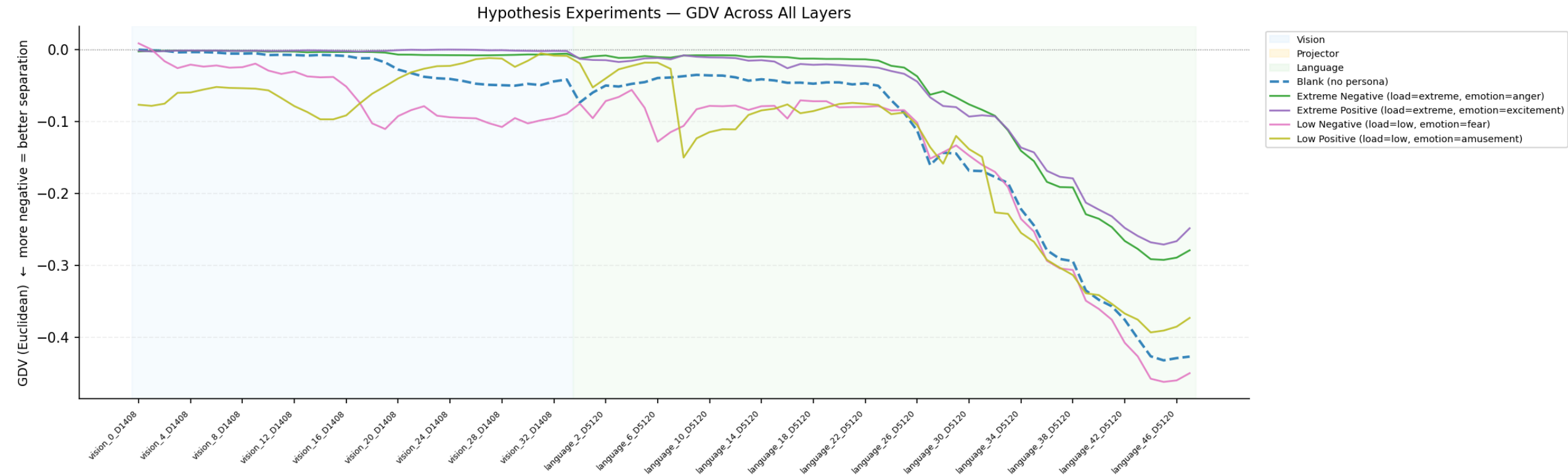
GDV Distribution Across All Layers — All Experiments
Box = IQR | Line = median | Whiskers = 1.5×IQR | Dots = outliers



Interestingness Label Distribution — All Experiments



GDV updated for Hypotheses



GDV updated for Hypotheses

Condition	Best GDV	Best Layer
Blank (no persona)	-0.4325	Language 45
Extreme Negative (load=extreme, emotion=anger)	-0.2927	Language 45
Extreme Positive (load=extreme, emotion=excitement)	-0.2713	Language 45
Low Negative (load=low, emotion=fear)	-0.4624	Language 45
Low Positive (load=low, emotion=amusement)	-0.3936	Language 44

Synonym Test: Extreme is a confounder?

Testing the label distribution with a changed prompt.

“extreme” mental load => “overwhelming” mental load

Why that choice?

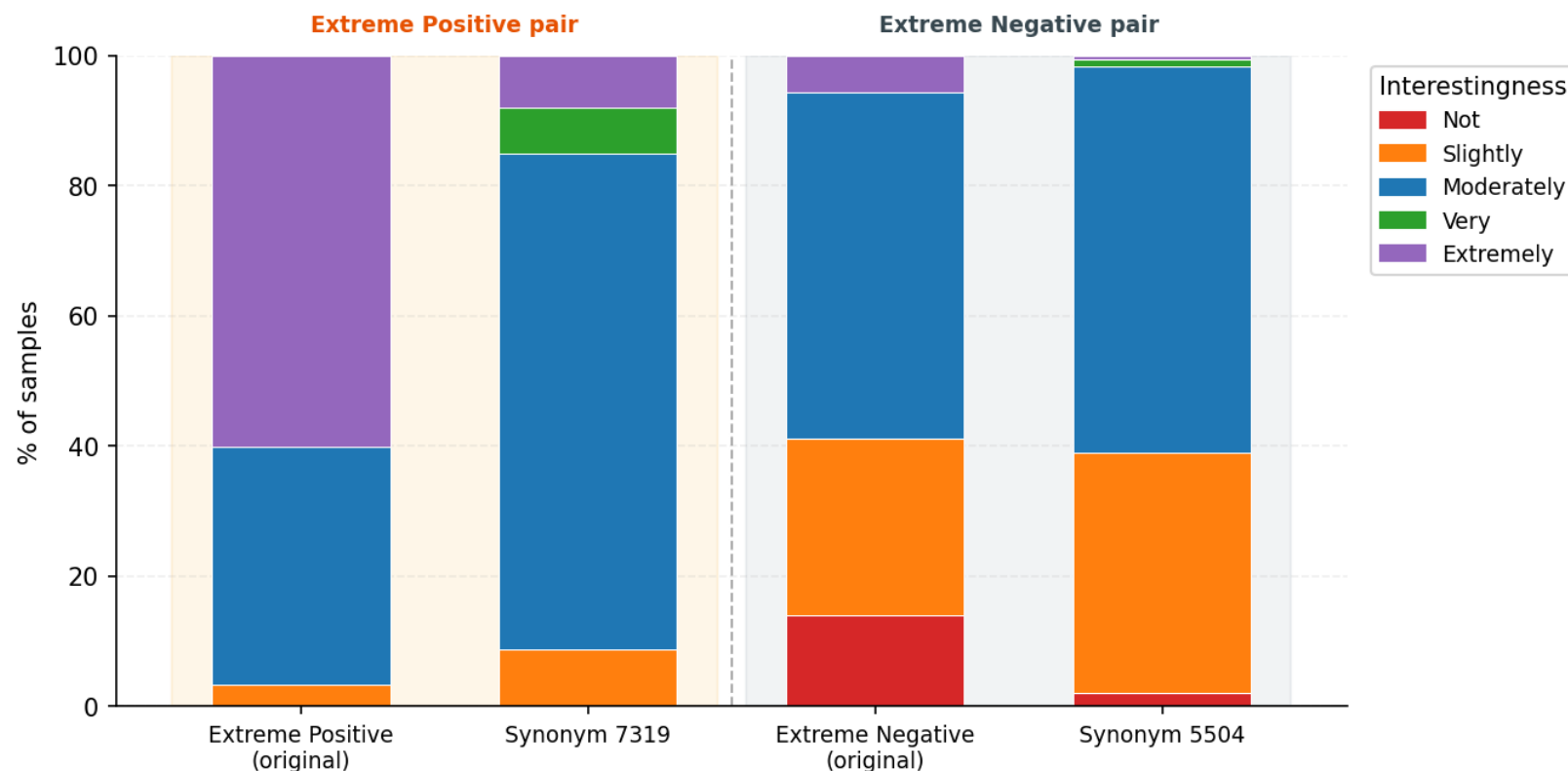
Mental load from low -> extreme has a psychological implication: psychologically speaking an extreme mental load is unsustainable and thereby has a negative connotation.

(would like to test “immense” as a less “negative-biased” synonym)

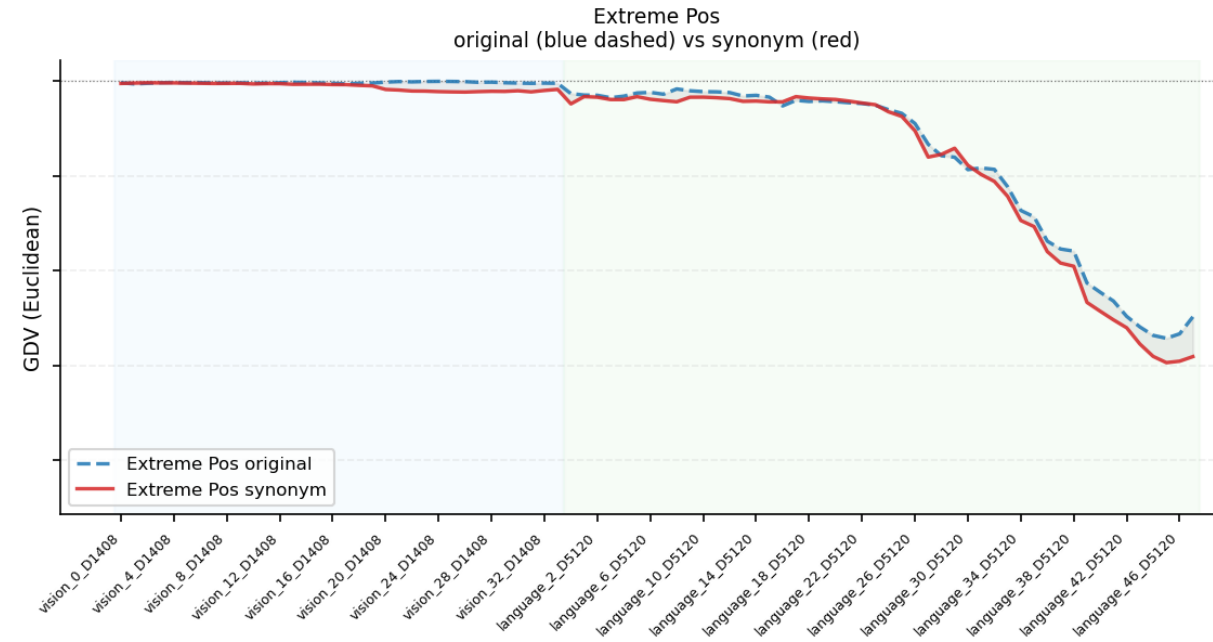
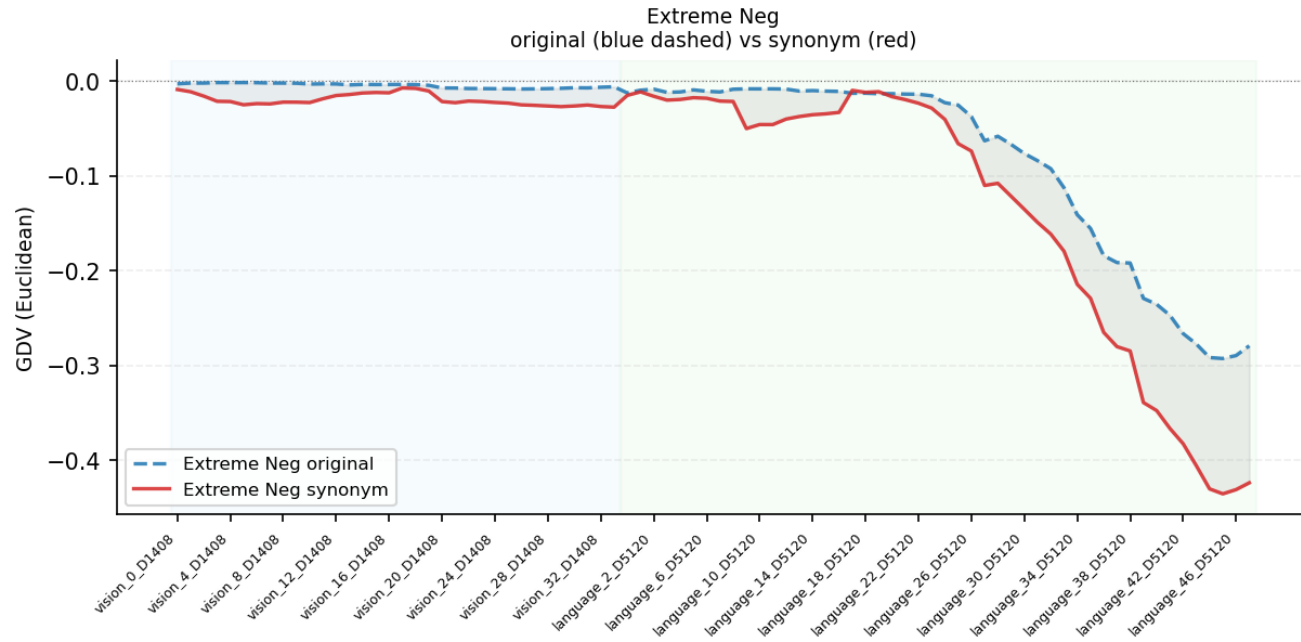
Extreme is a confounder?

Interestingness Label Distribution — Original vs Synonym Personas

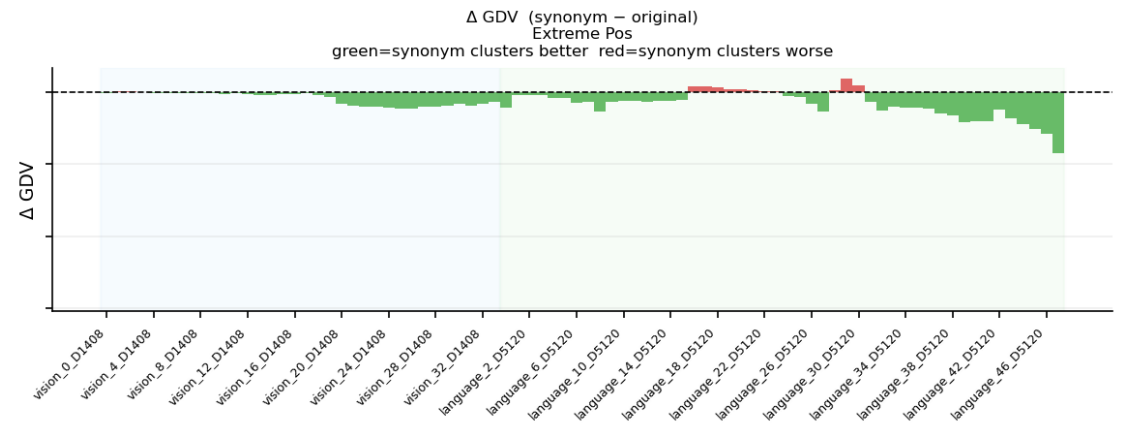
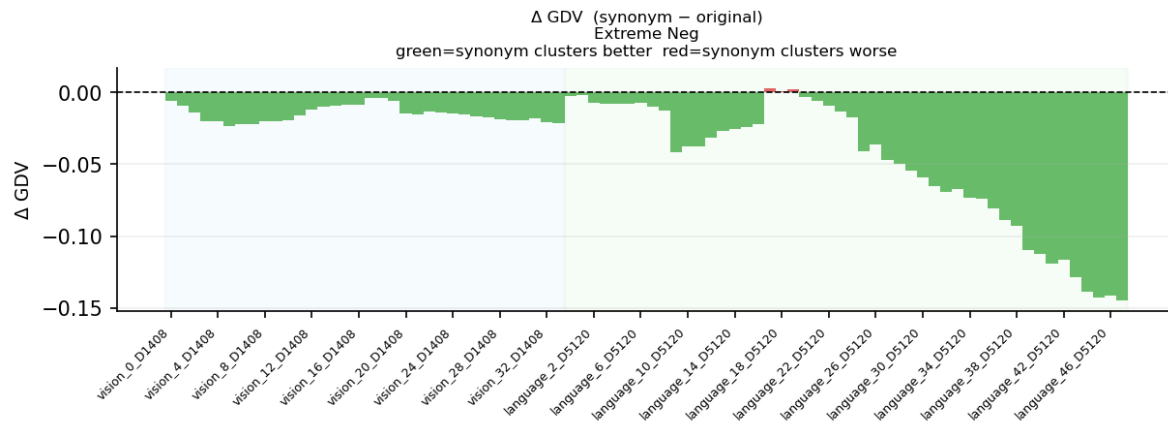
Condition	Best GDV	Best Layer	Δ best GDV vs original
Extreme Neg original	-0.2927	Language 45	
Extreme Neg synonym	-0.4355	Language 45	-0.1428
Extreme Pos original	-0.2713	Language 45	
Extreme Pos synonym	-0.2970	Language 45	-0.0257



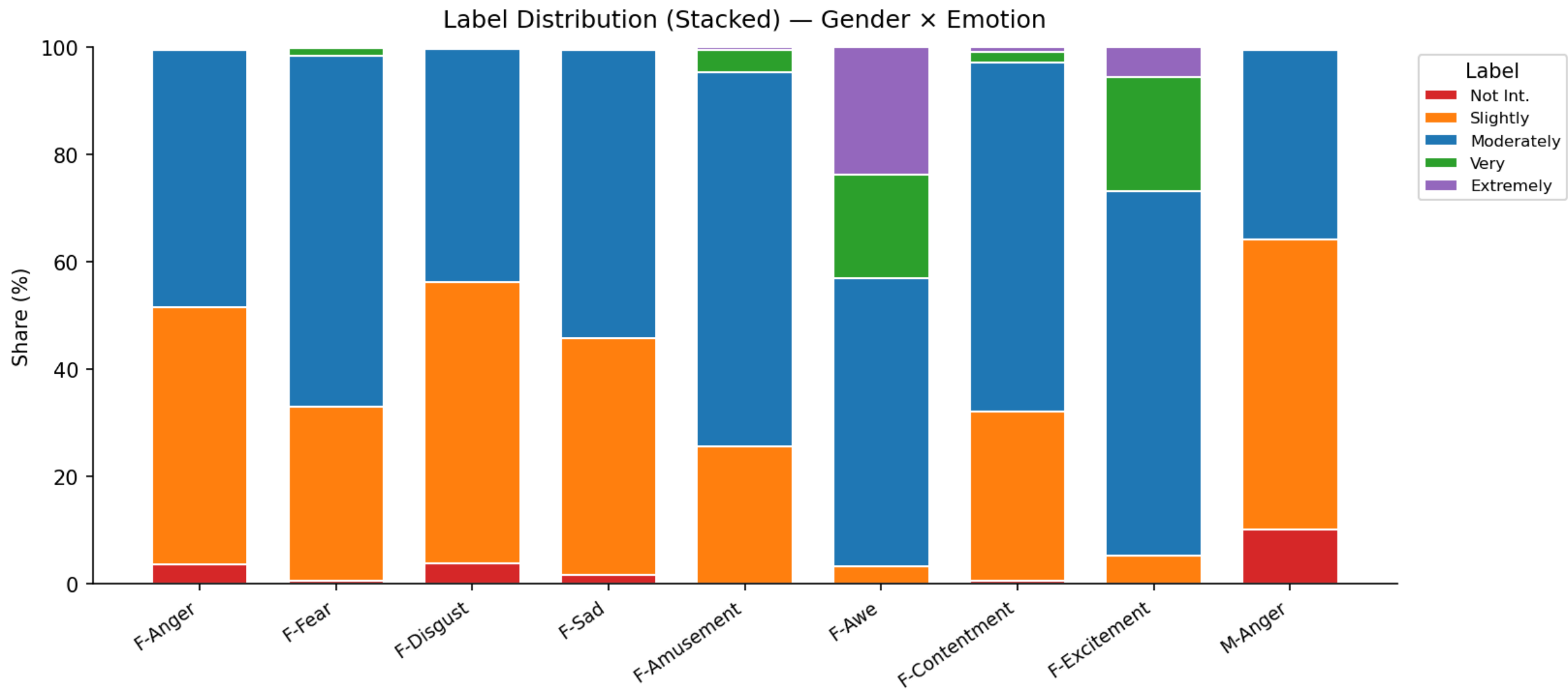
Synonym Test — Does Word Choice Shift Label Separation?



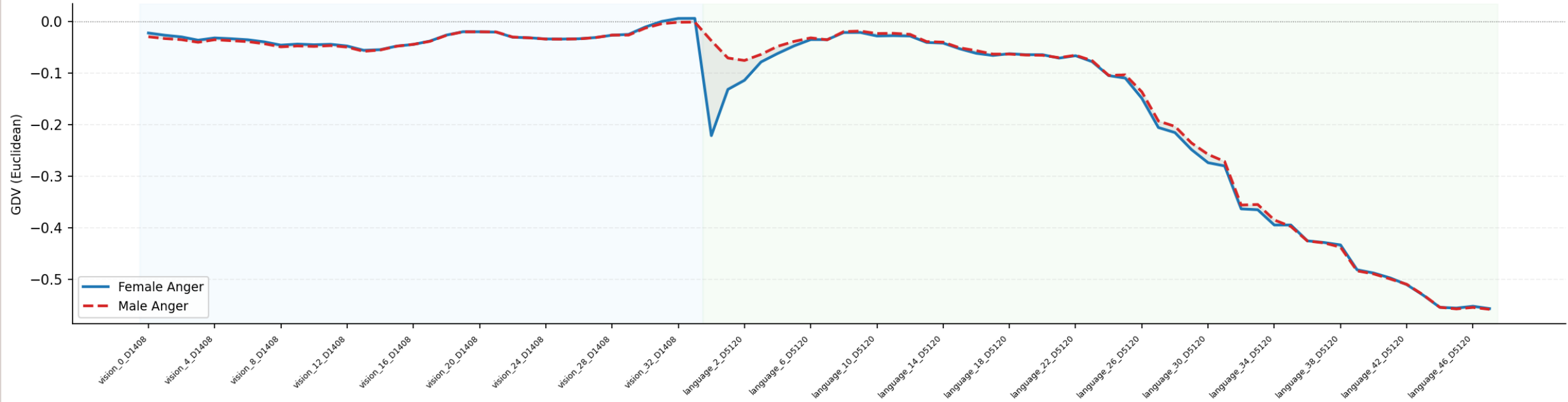
Synonym Effect — Δ GDV per Layer



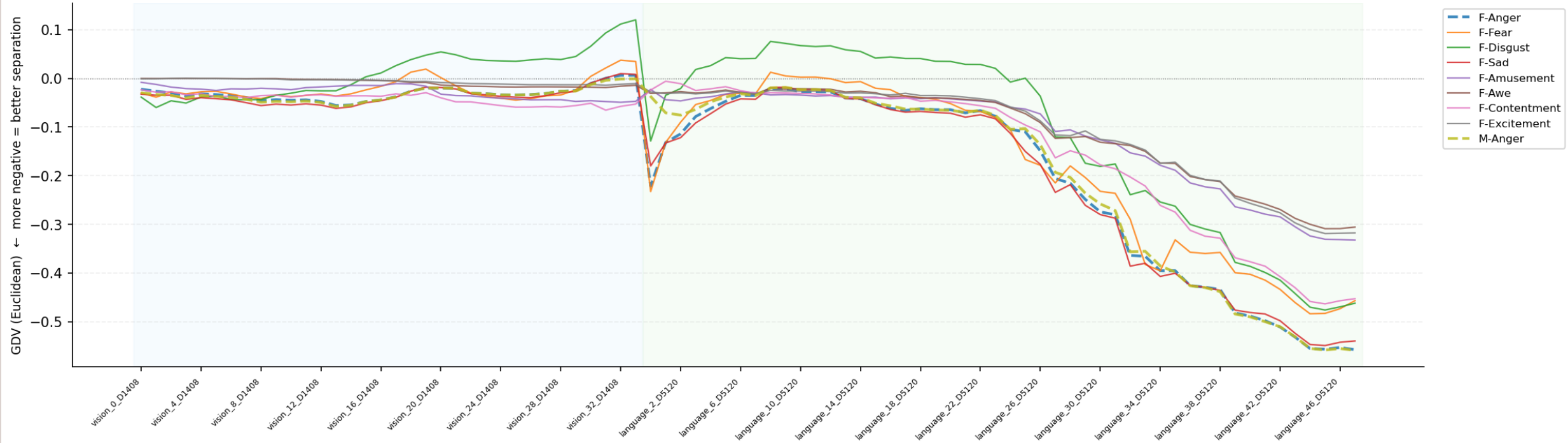
Impact of Gender vs. Emotion



Gender Effect — Female Anger vs Male Anger (same emotion, same images)



Gender x Emotion — GDV Across All Layers



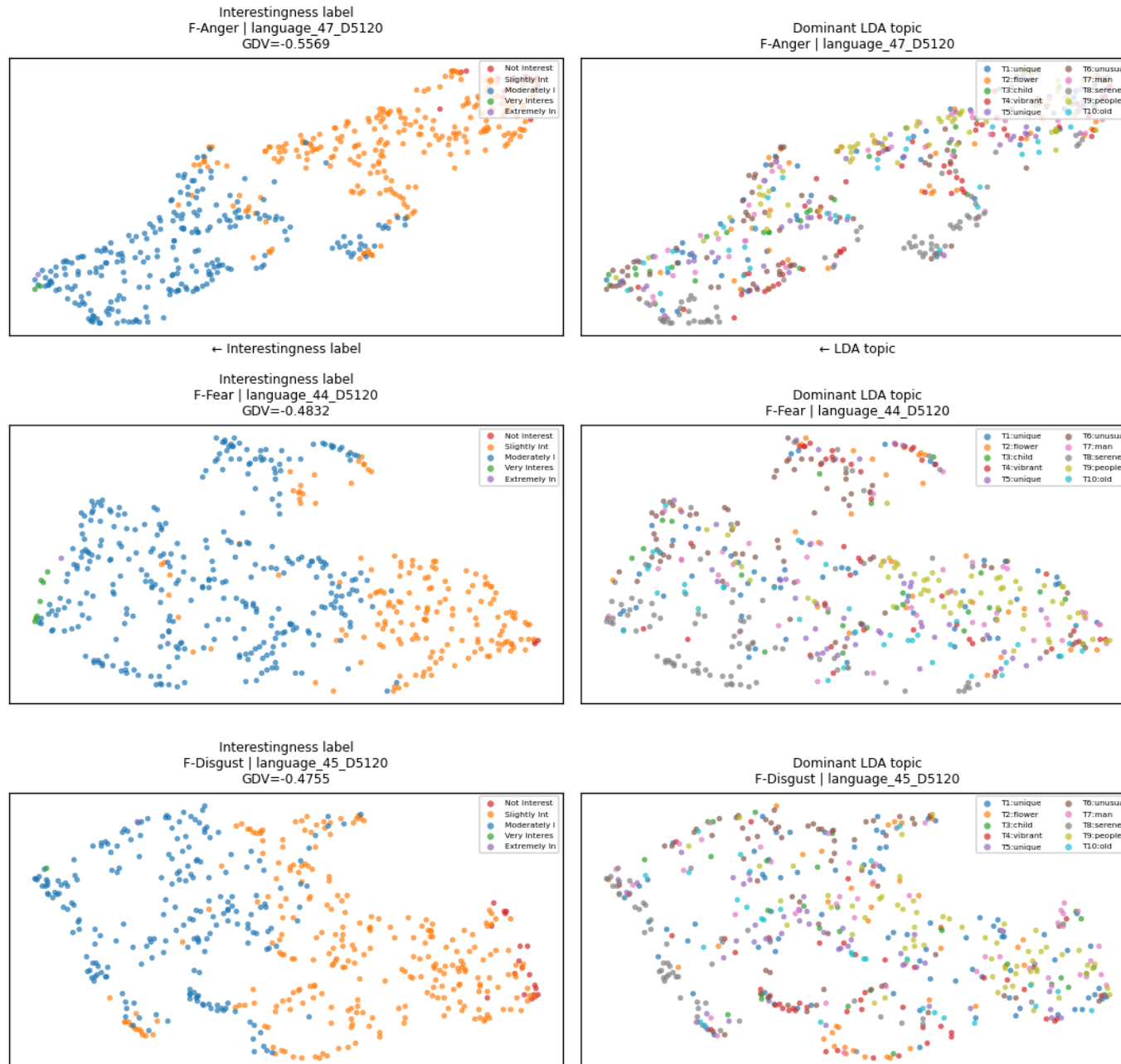
Impact of Gender vs. Emotion

Condition	Best GDV	Best Layer
F-Anger	-0.5569	Language 47
F-Fear	-0.4832	Language 44
F-Disgust	-0.4755	Language 45
F-Sad	-0.5484	Language 45
F-Amusement	-0.3320	Language 47
F-Awe	-0.3086	Language 45
F-Contentment	-0.4629	Language 45
F-Excitement	-0.3185	Language 45
M-Anger	-0.5583	Language 47

2 Ways to identify thematic clusters

1. Analyse the “explanations” via LDA / TF-IDF and color the dots on the gdv plots -> do they cluster?
2. Analyse the clusters on the gdv plot and ex-post try to identify visual or explanation-based unifying characteristics

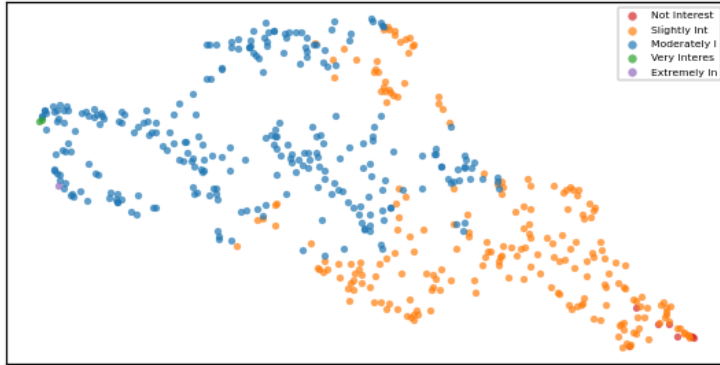
Label-coloured vs Topic-coloured Point Clouds — Best Overall Layer



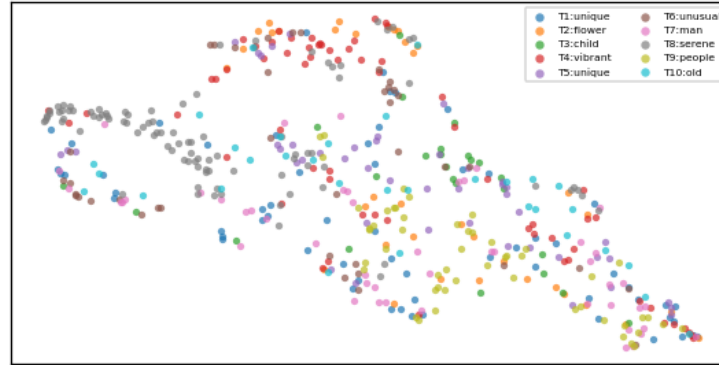
1. Analyse the “explanations” via LDA / TF-IDF and color the dots on the gdv plots -> do they they cluster?

- Thematic Areas exist but no clear thematic clustering by the “LDA” topics
- especially “serene” -> correlates with inter-subjective interestingness!
- “people” also somewhat clusters
- LDA might not be ideal for identifying “themes” from samples

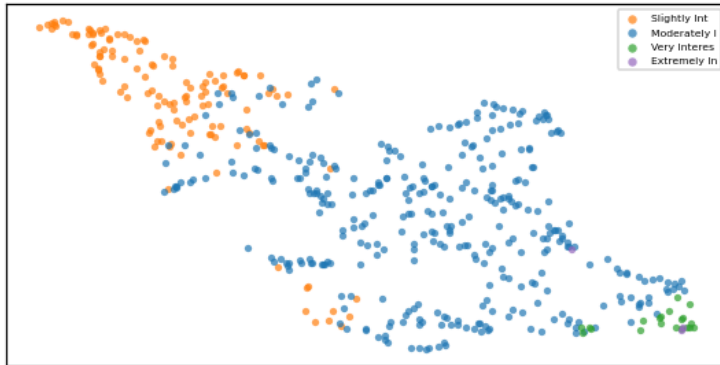
Interestingness label
F-Sad | language_45_D5120
GDV=-0.5484



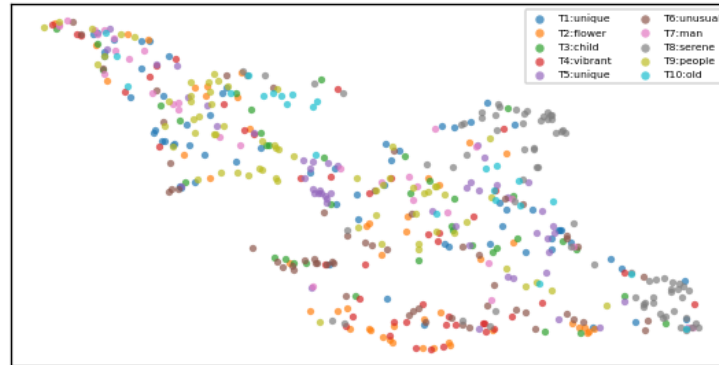
Dominant LDA topic
F-Sad | language_45_D5120



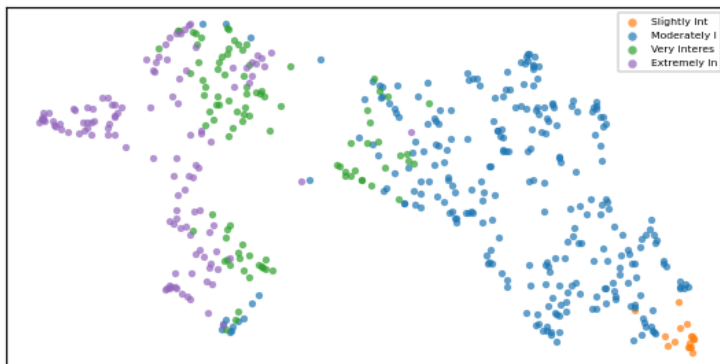
Interestingness label
F-Amus | language_47_D5120
GDV=-0.3320



Dominant LDA topic
F-Amus | language_47_D5120



Interestingness label
F-Awe | language_45_D5120
GDV=-0.3086

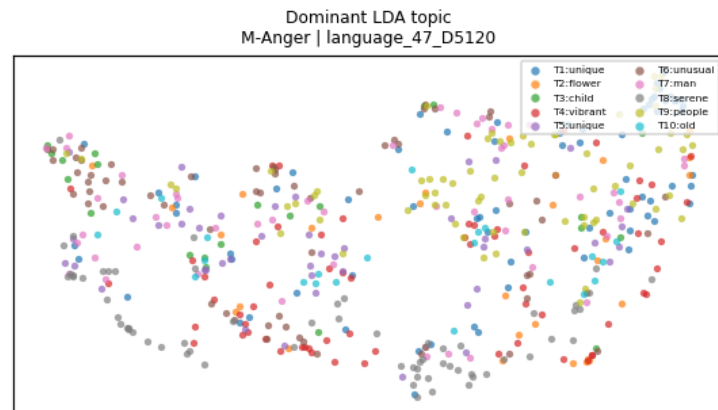
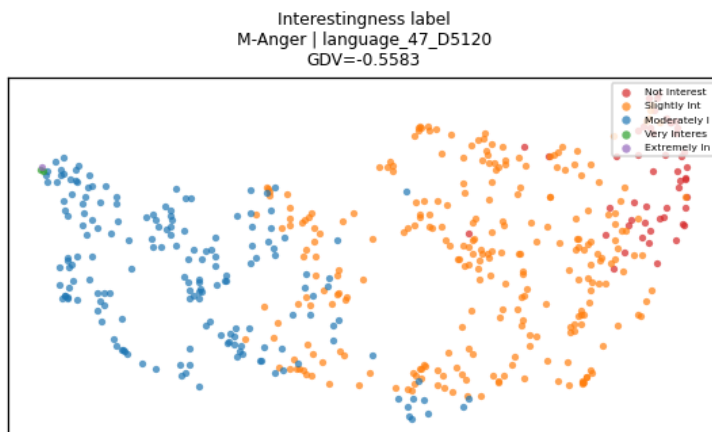
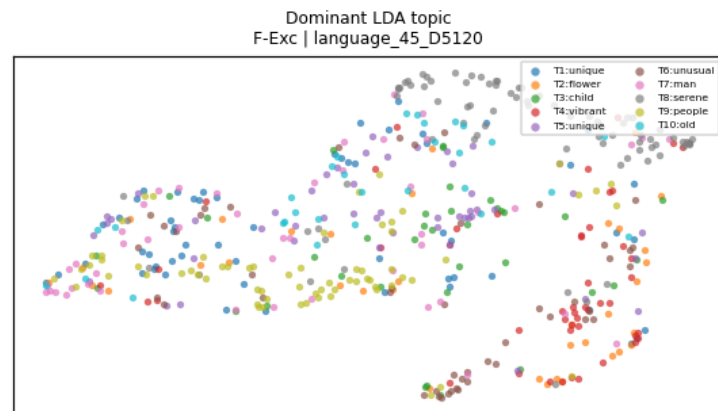
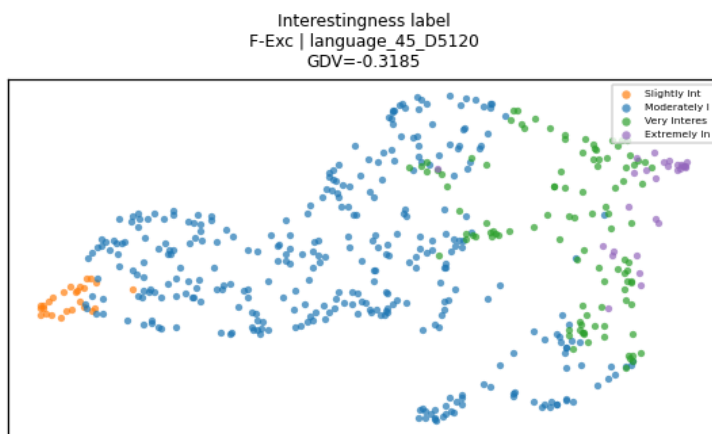
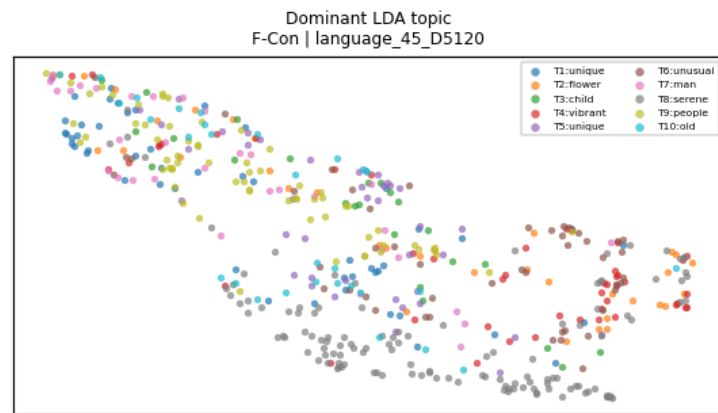
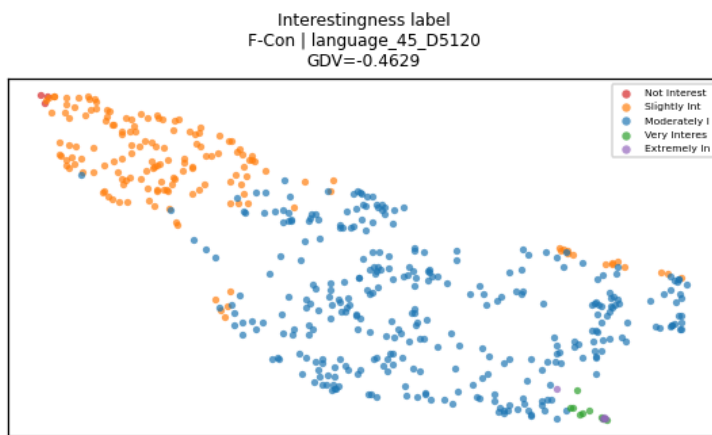


Dominant LDA topic
F-Awe | language_45_D5120



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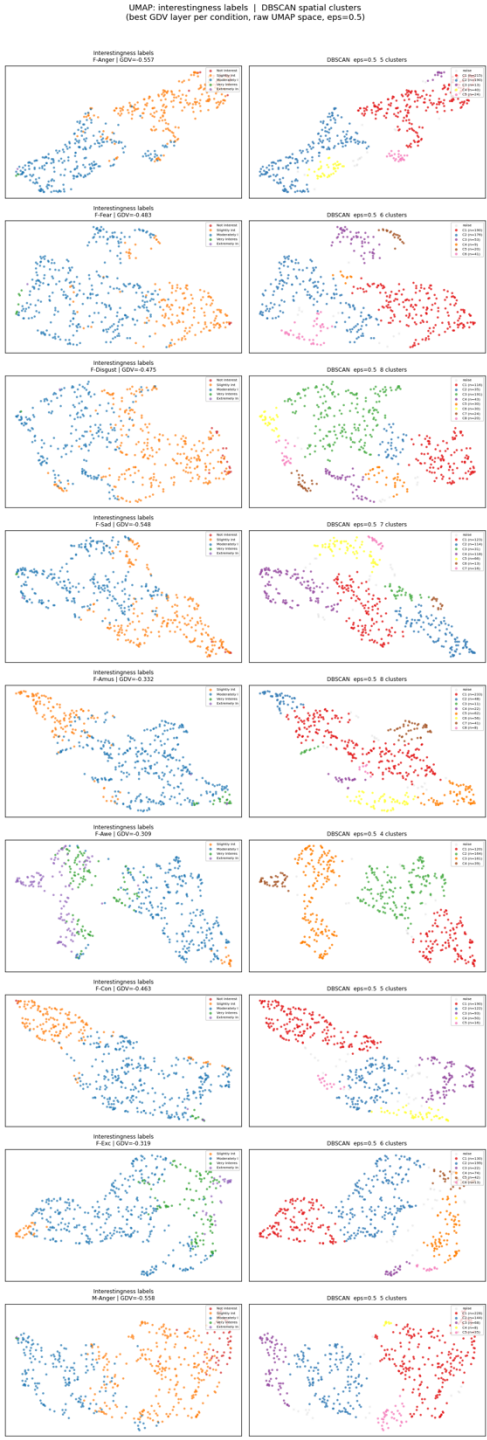
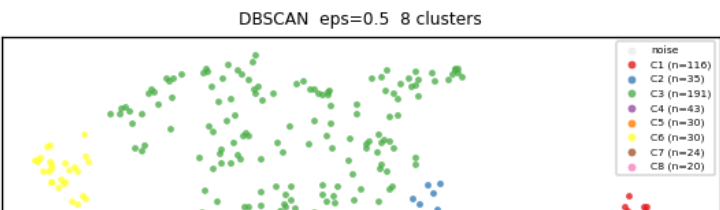
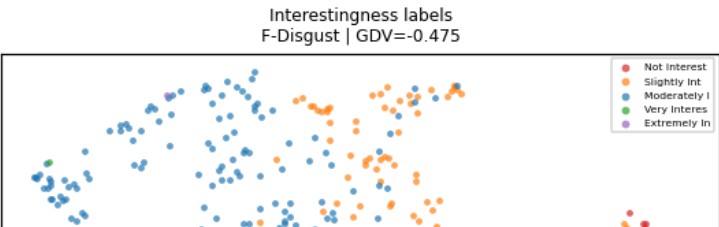
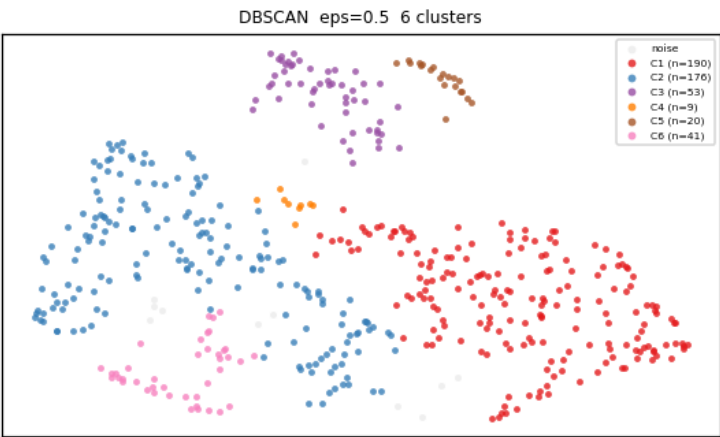
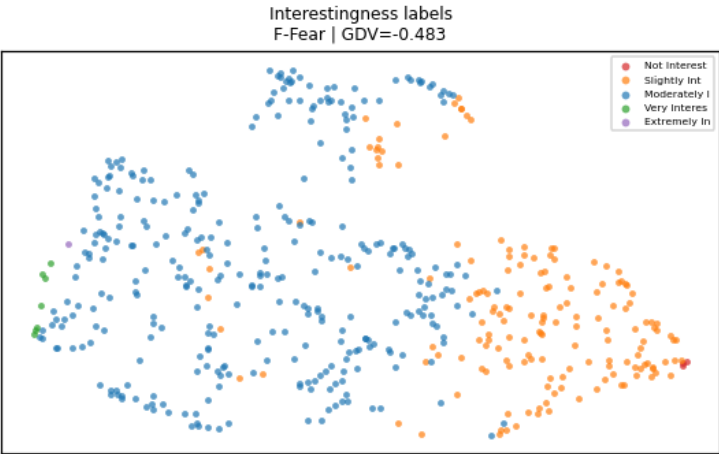
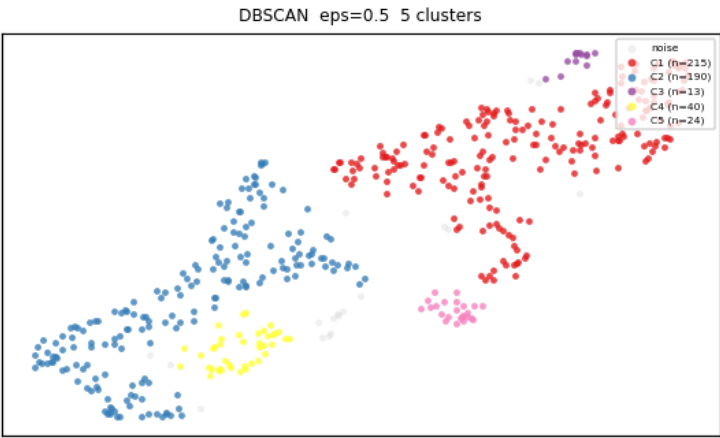
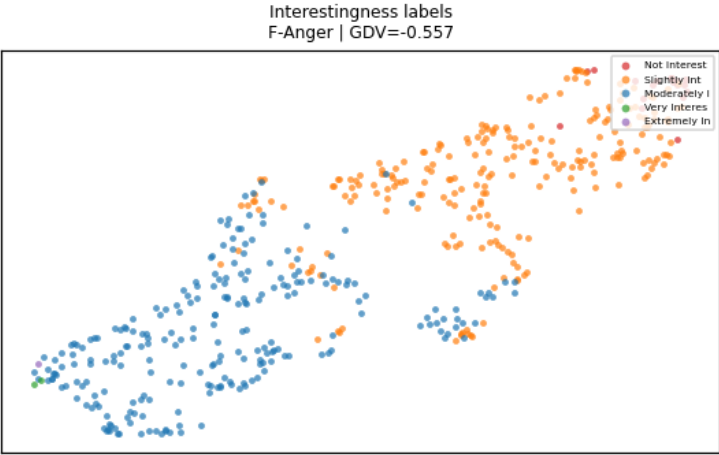


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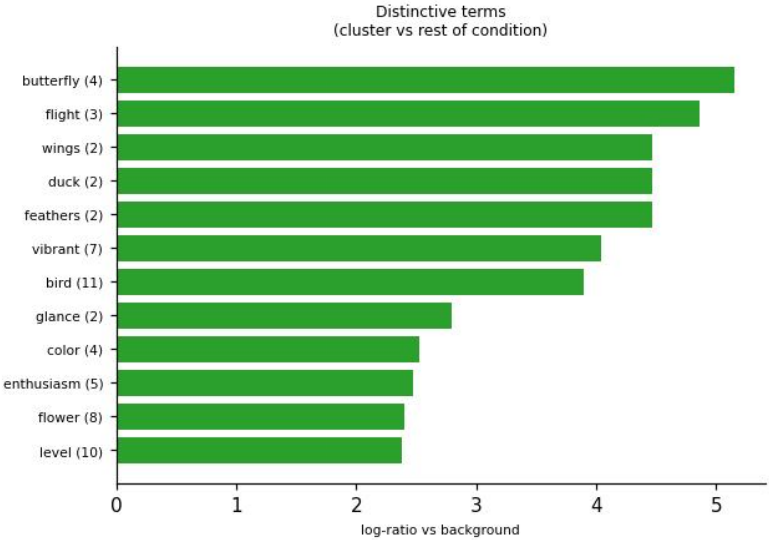
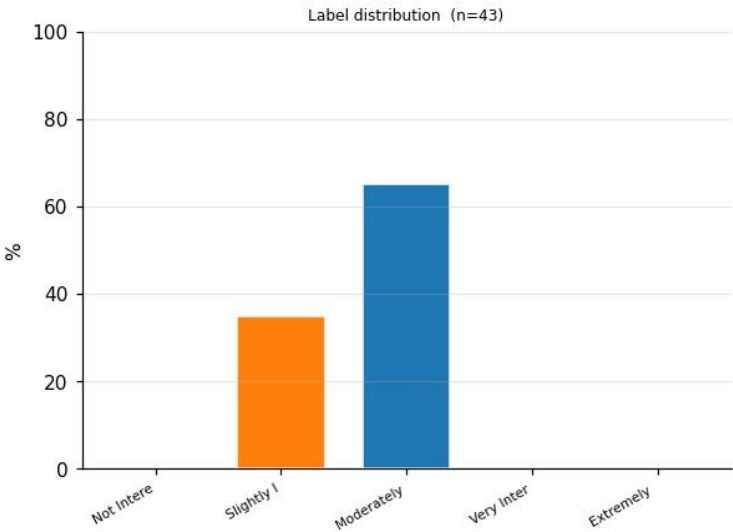
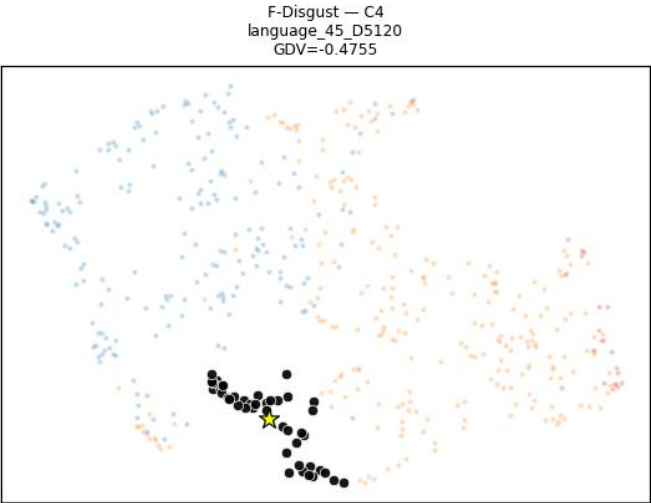
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2. ex-post identifying characteristics from existing clusters

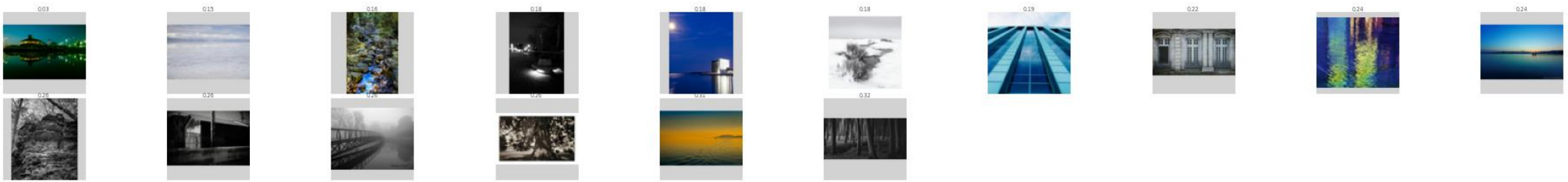
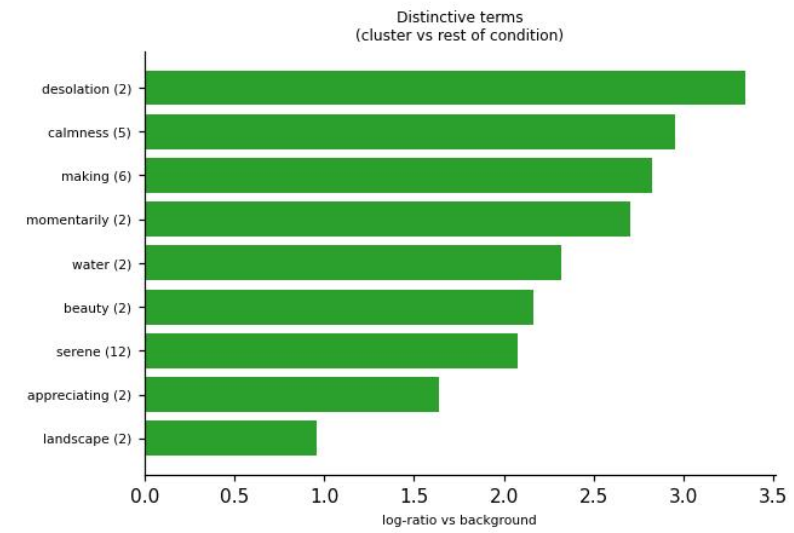
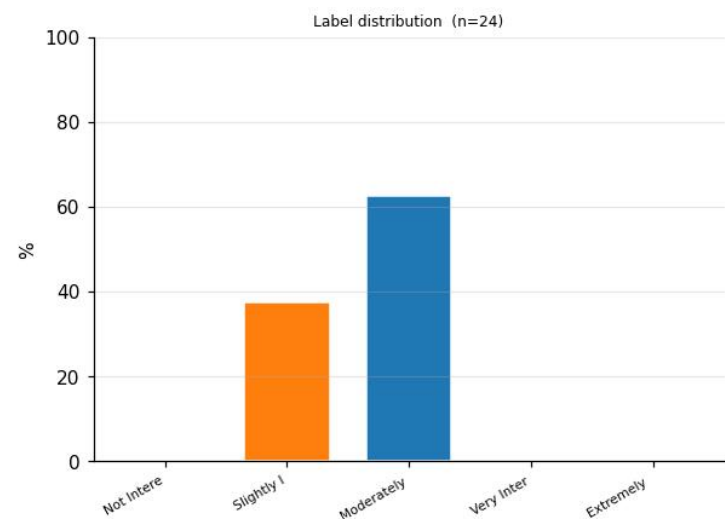
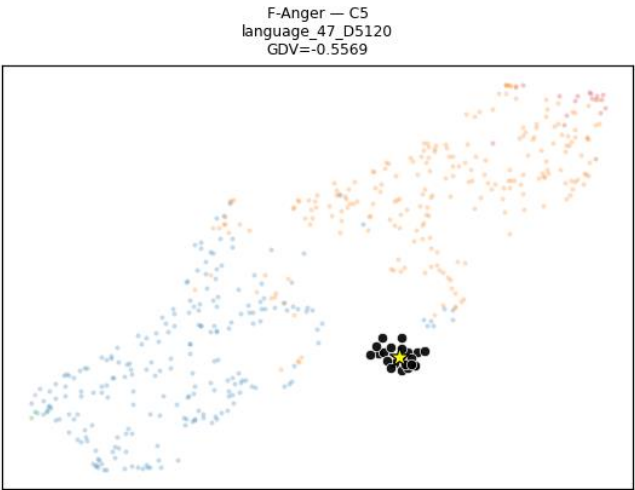
UMAP: interestingness labels | DBSCAN spatial clusters
(best GDV layer per condition, raw UMAP space, eps=0.5)



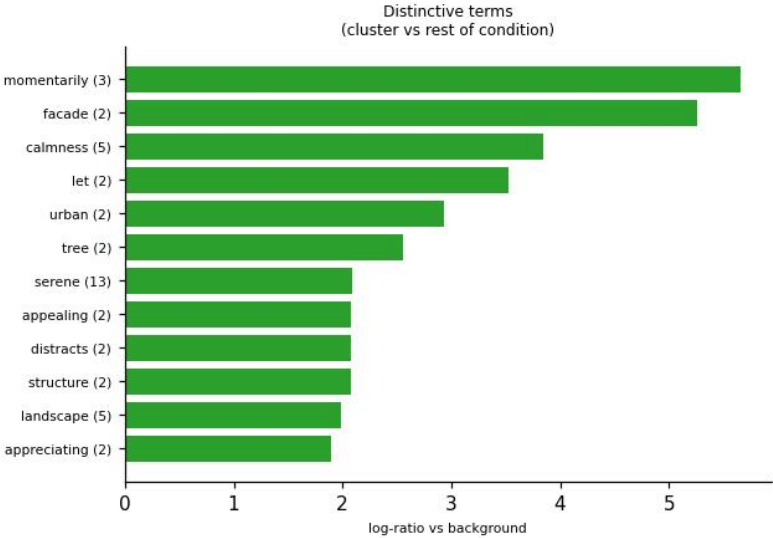
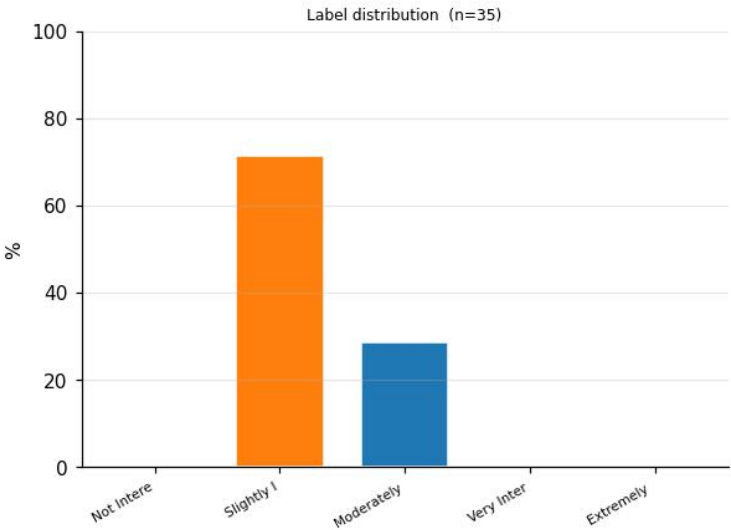
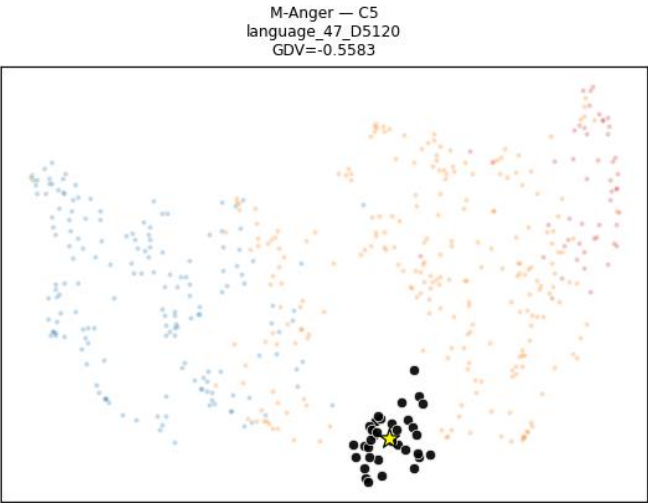
Spatial Cluster C4 — F-Disgust — 43 images (gallery sorted by centrality)



Spatial Cluster C5 — F-Anger — 24 images (gallery sorted by centrality)



Spatial Cluster C5 — M-Anger — 35 images (gallery sorted by centrality)



Most spatially stable images — same cluster regardless of emotion condition

0001.j



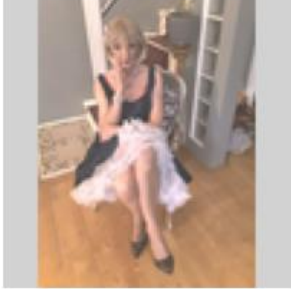
0011.j



0165.j



0311.j



0321.j



0809.j



0007.j



0013.j



0017.j



0019.j



0021.j



0025.j



0033.j



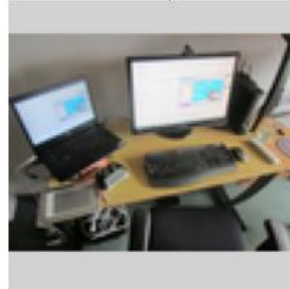
0035.j



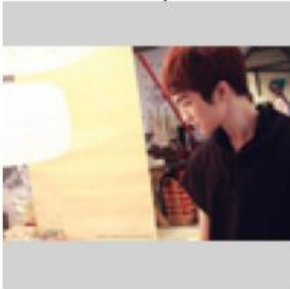
0063.j



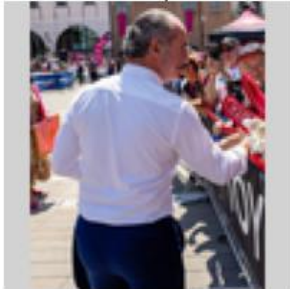
0107.j



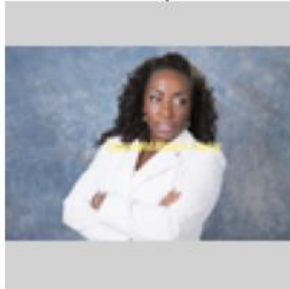
0117.j



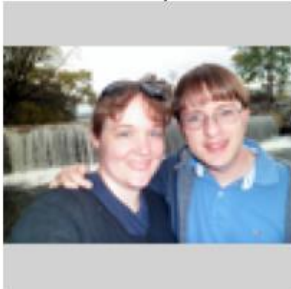
0143.j



0149.j



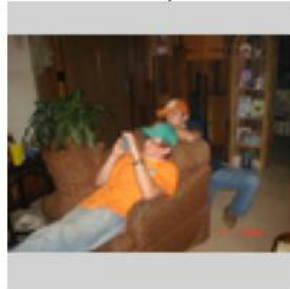
0161.j



0163.j



0169.j



0179.j



0185.j

